

2.23

a

$$P(1N) = 0.95^1$$

$$P(7N) = 0.95^7 = 0.698337$$

b

September has 30 days.

$$P(Sept = 2) = \binom{30}{2}(0.05)^2(0.95)^{28} = 0.2586367$$

c

$$10N = \{10 \text{ Days of no accidents}\}$$

Therefore at least 1 accident in 10 days is $10N^c$

$P(10N^c|4N) = 1 - P(6N)$ because we know that the accident must occur within the last 6 days

$$\rightarrow P(10N^c|4N) = 1 - (0.95^6) = 0.2469$$

$$P(10N^c \cap 4N) = P(10N^c|4N) * P(4N) = 0.2469 * 0.95^4 = 0.215769$$

2.28

a

Hypergeometric distribution, $X \text{ Hypergeom}(52, 4, 13)$

b

Binomial distribution, $X \text{ bin}(50, 1 - \frac{48!39!}{35!52!}) \rightarrow X \text{ bin}(50, \frac{14498}{20825})$

c

Binomial distribution, $X \text{ bin}(50, \frac{12!39!}{51!}) \rightarrow X \text{ bin}(50, \frac{1}{8255176274800})$

d

Hypergeometric distribution, $X \sim \text{Hypergeom}(52, 13, 13)$

2.47

$$P(2F) = \frac{55}{80} \frac{54}{79} = \frac{297}{632}$$

2.66

We can treat this problem as a binomial distribution with $X \sim \text{bin}(n, \frac{1}{6})$ and we are solving for an n with the highest resulting probability.

$$P(X = 2) = \binom{n}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^{n-2} \text{ for } n = 1, 2, 3, \dots, 50$$

I will use the python code below to solve for n :

```
from math import comb
bestProb = 0
bestRolls = 0
for n in range(2,40):
    prob=(comb(n,2) * ((1/6)**2) * ((5/6)**(n-2)))
    if prob > bestProb:
        bestProb = prob
        bestRolls = n
print(f"Optimal number of rolls is {bestRolls} resulting in a probability of {bestProb}")
```

Optimal number of rolls is 11 resulting in a probability of 0.2960935686313839

Therefore the optimal number of rolls is actually 11 resulting in a probability of 0.296

2.71

a

b

c

d

2.81